

Time Series

Spring Semester 2026, EPFL
École Polytechnique Fédérale de Lausanne
Prof. Dr. Sofia C. Olhede

Practice Exam — Week 1: Stationarity, Autocovariance & Ergodicity

Duration: 60 minutes · No calculator · Answer on paper

Name: _____ Surname: _____

SCIPER (Student Card Number): _____

Exercise	Points
1	
2	
3	
4	
Total	

Justify every answer. Throughout, $\{\varepsilon_t\}$ denotes a mean-zero white noise of variance σ^2 .

Problem 1 (5 points)

(a) State the definition of *weak (second-order) stationarity* and of *strict stationarity* for a process $\{X_t\}_{t \in \mathbb{Z}}$. (2 pts)

(b) Let $Y_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_Y^2)$ and let a, b, c be constants. For each process, decide whether it is weakly and/or strictly stationary, and give its mean and ACVS:

(i) $X_t = a + bY_t + cY_{t-1}$;

(ii) $X_t = Y_0 \cos(ct)$.

(2 pts)

(c) Give a process that is strictly stationary but *not* weakly stationary, and explain why. (1 pt)

Problem 2 (5 points)

Let $\{X_t\}$ be a second-order stationary process with ACVS γ_τ , *independent* of the white noise $\{\varepsilon_t\}$.

(a) Determine the ACVS of $Z_t = X_t + \varepsilon_t$. (2 pts)

(b) Let $W_t = \theta_1 \varepsilon_t + \theta_2 \varepsilon_{t-1}$. Compute its ACVS γ_τ^W for all τ and its autocorrelation ρ_1 , and prove that $|\rho_1| \leq \frac{1}{2}$. (3 pts)

Problem 3 (5 points)

(a) Prove that the autocovariance sequence $\{\gamma_\tau\}$ of a stationary process is *positive semi-definite*: for every $n \geq 1$ and all $a_1, \dots, a_n \in \mathbb{R}$,

$$\sum_{j=1}^n \sum_{k=1}^n a_j a_k \gamma_{j-k} \geq 0.$$

(2 pts)

(b) Deduce the symmetry $\gamma_{-\tau} = \gamma_\tau$ and the bound $|\gamma_\tau| \leq \gamma_0$.

(1 pt)

(c) Is the sequence $\gamma_0 = 1$, $\gamma_{\pm 1} = 0.8$, $\gamma_\tau = 0$ ($|\tau| \geq 2$) a valid ACVS of some stationary process? Justify.

(2 pts)

Problem 4 (5 points)

For observations $X_1, \dots, X_n$ recall the two sample-autocovariance estimators (for $|\tau| \leq n-1$, and 0 otherwise):

$$\tilde{\gamma}_\tau = \frac{1}{n-|\tau|} \sum_{t=1}^{n-|\tau|} (X_t - \bar{X})(X_{t+|\tau|} - \bar{X}), \quad \hat{\gamma}_\tau = \frac{1}{n} \sum_{t=1}^{n-|\tau|} (X_t - \bar{X})(X_{t+|\tau|} - \bar{X}).$$

- (a) Which estimator is biased? Which one is positive semi-definite? (1 pt)
- (b) For Gaussian white noise (known mean) one has, for $\tau \neq 0$, $\text{Var}(\hat{\gamma}_\tau) = \frac{n-|\tau|}{n^2} \sigma^4$ and $\text{Var}(\tilde{\gamma}_\tau) = \frac{1}{n-|\tau|} \sigma^4$. Which estimator has the smaller mean-squared error, and why? (2 pts)
- (c) Show that if $\sum_\tau |\gamma_\tau| < \infty$ then $\text{Var}(\bar{X}) \rightarrow 0$ as $n \rightarrow \infty$, so $\bar{X}$ is a consistent (mean-ergodic) estimator of $\mu = \mathbb{E}[X_t]$. (2 pts)

